Hotel Booking Cancellation Prediction Using Supervised Learning Classifiers and Unsupervised K-Means Clustering

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***Abstract*—Hotel booking cancellations represent a major operational and financial challenge in the hospitality industry, causing severe revenue loss, room vacancies, and unpredictable resource allocation. This study develops a comprehensive machine learning system to predict whether a reservation will be cancelled (`is_canceled` = 1) or honored (`is_canceled` = 0) at the time of booking. Using the benchmark Hotel Booking Demand dataset (119,390 records and 32 attributes), we first identify and remove post-booking data leakage features (`reservation_status` and `reservation_status_date`), handle high-cardinality nominals, impute missing values, and execute stratified 80/20 partitioning. Five supervised classifiers are trained and systematically evaluated: K-Nearest Neighbors (KNN), Decision Tree, Logistic Regression, Gaussian Naive Bayes, and a Deep Neural Network (DNN) with Batch Normalization and Dropout regularization. In addition, an unsupervised K-Means clustering analysis ($k = 2$) is conducted to explore natural data partitioning against ground truth labels. The Deep Neural Network achieves the highest overall predictive performance with an accuracy of 81.51%, an F1-score of 0.6223, and an AUC-ROC of 0.8585, closely followed by the Decision Tree (80.39% accuracy, 0.8305 AUC).**

***Index Terms*—Hotel booking cancellation, binary classification, data leakage, exploratory data analysis, neural networks, decision trees, K-Means clustering, ROC analysis, revenue management.**

I. Introduction

THE global hospitality sector operates under perishable inventory constraints: once a night passes with an unoccupied room, the opportunity to generate that revenue is permanently lost. Unanticipated reservation cancellations, which frequently exceed 35% to 40% in commercial hotels, force managers into a difficult trade-off between accepting high vacancy losses and implementing aggressive overbooking policies that risk guest dissatisfaction [1].

Machine learning provides a powerful predictive mechanism to estimate cancellation probability at the instant a reservation is placed. By identifying high-risk bookings in advance, hoteliers can deploy targeted customer retention campaigns, calibrate optimal deposit policies, and dynamically regulate room allocations [4].

In this project, we formulate the problem as a *supervised binary classification* task. We select the discrete attribute `is_canceled` $\in \{0, 1\}$ as our primary target variable. Given an input feature vector $\mathbf{x} \in \mathbb{R}^d$, the goal is to learn a decision boundary $f(\mathbf{x}) \to \hat{y}$ that separates confirmed stays ($\hat{y} = 0$) from cancellations ($\hat{y} = 1$).

The core contributions of this work are fourfold:

1) **Data Leakage Eradication:** Detecting and stripping outcome-revealing post-booking fields (`reservation_status`, `reservation_status_date`) that falsely inflate test metrics.
2) **Exploratory Data Analysis:** Comprehensive statistical evaluation using Pearson correlation, outlier boxplots, and categorical cancellation rates.
3) **Supervised Benchmark:** Rigorous evaluation of five distinct machine learning models under stratified validation splits.
4) **Unsupervised Structure Discovery:** K-Means clustering ($k = 2$) validated via the Elbow Method (WCSS) and PCA 2-D visual projections.

II. Dataset Description

A. Problem Formulation and Dataset Scale

This project is formulated as a **binary classification problem** because the objective is to predict a discrete categorical outcome with exactly two states: $y = 0$ (Not Cancelled / Check-Out) and $y = 1$ (Cancelled / No-Show). A regression approach is unsuitable here because the ground truth labels are discrete classes rather than continuous numerical quantities.

The dataset (`11.csv`) comprises $N = 119{,}390$ individual booking observations spanning 32 attributes recorded between July 2015 and August 2017 across Resort and City hotel properties in Portugal [1].

B. Feature Typology

The 32 features are categorized into two primary statistical types:

- **Quantitative Features (Numerical):** Discrete integer counts and continuous metrics, including `lead_time`, `stays_in_weekend_nights`, `stays_in_week_nights`, `adults`, `children`, `babies`, `previous_cancellations`, `previous_bookings_not_canceled`, `booking_changes`, `days_in_waiting_list`, `adr` (Average Daily Rate), `required_car_parking_spaces`, and `total_of_special_requests`.

Both authors are undergraduate students in the Department of Computer Science and Engineering, BRAC University, Dhaka, Bangladesh. Course: CSE422 — Artificial Intelligence Laboratory, Summer 2026. Group: 11. Dataset: `11.csv`.

- **Qualitative Features (Categorical):** Nominal identifiers and categories, including `hotel`, `arrival_date_month`, `meal`, `country`, `market_segment`, `distribution_channel`, `reserved_room_type`, `assigned_room_type`, `deposit_type`, `customer_type`, `reservation_status`, and `reservation_status_date`.

C. Dataset Class Balance

Evaluating the output target feature (`is_canceled`) reveals a mild-to-moderate class imbalance:

- **Class 0 (Not Cancelled):** 75,166 bookings ($\approx 62.96\%$).
- **Class 1 (Cancelled):** 44,224 bookings ($\approx 37.04\%$).

Figure 1 displays the bar chart of the two target classes, illustrating the exact frequency and relative proportions.

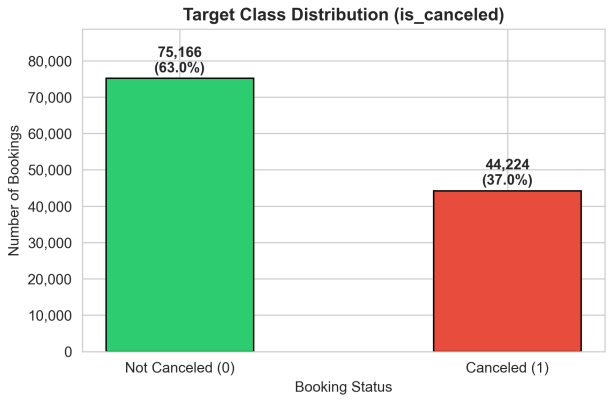


Fig. 1. Class balance distribution of the target variable `is_canceled`.

III. Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to uncover key structural patterns and relationships governing cancellation behavior.

A. Key Relationships from EDA

- **Lead Time Distribution:** The histogram of `lead_time` displays a marked right skew. Guests booking several months in advance exhibit substantially higher cancellation frequencies than short-notice reservations.
- **Special Requests Impact:** Analysis reveals an inverse relationship between `total_of_special_requests` and cancellations. Guests requesting specific amenities (e.g., high floor, twin beds) demonstrate higher commitment to their travel plans.
- **Deposit Policy Effect:** Crosstab analysis demonstrates that bookings with a *Non-Refund* deposit policy exhibit an extreme $\approx 99\%$ cancellation rate in this dataset, indicating that non-refundable rates were primarily chosen by high-risk group bookings.

B. Correlation Test Analysis

Figure 2 presents the Pearson linear correlation heatmap computed across all quantitative features.

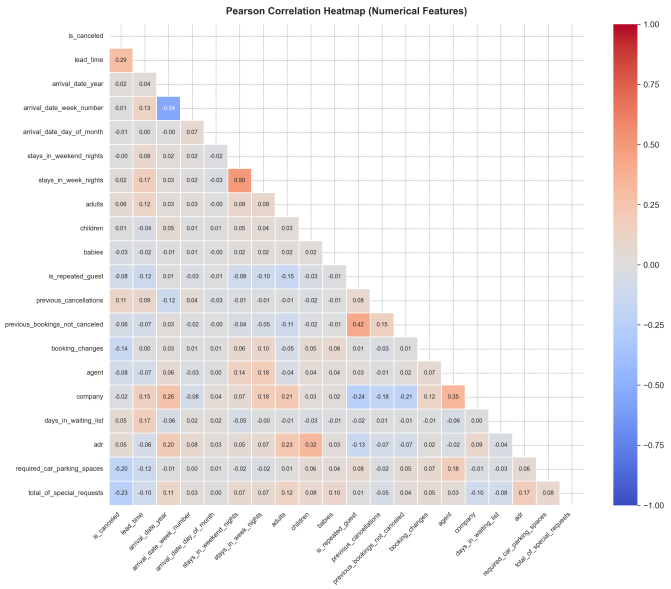


Fig. 2. Pearson correlation heatmap across quantitative dataset features.

The correlation analysis reveals:

1) Positive linear correlation with cancellation: `lead_time` ($r \approx +0.29$) and `previous_cancellations` ($r \approx +0.11$).
2) Negative linear correlation with cancellation: `total_of_special_requests` ($r \approx -0.23$) and `required_car_parking_spaces` ($r \approx -0.20$).
3) **Data Leakage Discovery:** The attribute `reservation_status` exhibited an absolute correlation ($r > 0.99$) with `is_canceled` because it directly logs the post-arrival status (*Canceled*, *Check-Out*, *No-Show*). Training on this column would yield an artificial 100% accuracy model that is entirely useless in production.

IV. Dataset Pre-Processing

To transform raw transactions into a reliable input matrix, we resolved several critical data issues:

A. Problem 1: Data Leakage Removal

Problem: `reservation_status` and `reservation_status_date` are recorded only after the booking outcome is finalized.
Solution: Both columns were permanently removed prior to any feature extraction.

B. Problem 2: High-Cardinality Feature Dimensionality

Problem: The nominal feature `country` contains 178 distinct categories, which would cause severe dimensional explosion under one-hot encoding or inject artificial ordinal distances under label encoding.
Solution: The `country` column was dropped from the feature space.

C. Problem 3: Missing Value Handling

Problem: Missing entries were present in `agent` (16,340 nulls), `company` (112,593 nulls), and `children` (4 nulls).
Solution: In domain context, a null in `agent` or `company` indicates direct individual bookings without agency sponsorship. These were imputed with 0. The 4 missing values in `children` were imputed with 0.

D. Problem 4: Duplicate Records

Problem: Identical booking entries over-represent specific patterns and risk overfitting.
Solution: Exact duplicate rows were filtered out via deduplication.

E. Problem 5: Categorical Encoding

Problem: Machine learning algorithms require numerical input matrices.
Solution: Scikit-learn's `LabelEncoder` was applied to map nominal features (`hotel`, `meal`, `market_segment`, `distribution_channel`, `reserved_room_type`, `assigned_room_type`, `deposit_type`, `customer_type`) into integer vectors.

F. Problem 6: Disparate Feature Scales

Problem: Distance-based (KNN) and gradient-based (Logistic Regression, Neural Networks) algorithms are biased toward high-magnitude features like `adr` or `lead_time`.
Solution: Standard Z-score normalization was applied using `StandardScaler`.

V. Dataset Splitting

A. Stratified Partitioning

To preserve the 62.96% to 37.04% class distribution in both training and testing phases, we employed an **80/20 Stratified Split** with a fixed random state (seed = 42):

- **Training Set ($\mathbf{X}_{\text{train}}, \mathbf{y}_{\text{train}}$):** 80% of observations ($\approx$ 95,100 samples).
- **Testing Set ($\mathbf{X}_{\text{test}}, \mathbf{y}_{\text{test}}$):** 20% of observations ($\approx$ 23,790 samples).

B. Feature Scaling Protocol

The `StandardScaler` was fitted exclusively on $\mathbf{X}_{\text{train}}$ and used to transform both $\mathbf{X}_{\text{train}}$ and $\mathbf{X}_{\text{test}}$:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{1}$$

This prevents data leakage by ensuring that no test distribution statistics inform the training process.

VI. Model Training and Testing (Supervised)

Five distinct classifiers spanning instance-based, tree-based, linear, probabilistic, and deep learning paradigms were trained and evaluated.

A. K-Nearest Neighbors (KNN)

KNN classifies query instances by taking a majority vote among their $k = 5$ nearest neighbors in standardized Euclidean space:

$$d(\mathbf{x}, \mathbf{z}) = \sqrt{\sum_{j=1}^{d} (x_j - z_j)^2} \tag{2}$$

B. Decision Tree Classifier

Decision trees construct non-linear decision boundaries by recursively selecting feature splits that maximize the reduction in Gini Impurity:

$$I_G(p) = 1 - \sum_{k=0}^{1} p_k^2 \tag{3}$$

Configuration: Gini criterion, max_depth = 10, min_samples_split = 20. Figure 3 displays the top 12 features ranked by Gini importance score.

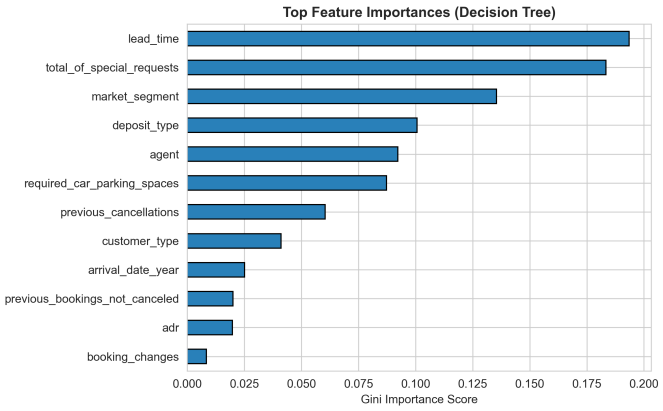


Fig. 3. Top 12 feature importances computed by the Decision Tree.

C. Logistic Regression

Logistic Regression estimates the posterior probability using the logistic sigmoid function:

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T\mathbf{x}+b)}} \tag{4}$$

Optimized using the L-BFGS solver with L_2 regularization (max_iter = 1000).

D. Gaussian Naive Bayes

Gaussian NB computes class posterior probabilities assuming feature conditional independence:

$$P(y = k|\mathbf{x}) \propto P(y = k) \prod_{j=1}^{d} \mathcal{N}(x_j; \mu_{kj}, \sigma_{kj}^2) \tag{5}$$

E. Deep Neural Network (DNN)

A 4-layer feedforward Deep Neural Network was designed:

- **Architecture:** Dense$(128, \text{ReLU}) \rightarrow$ BatchNorm $\rightarrow$ Dropout$(0.3) \rightarrow$ Dense$(64, \text{ReLU}) \rightarrow$ BatchNorm $\rightarrow$ Dropout$(0.2) \rightarrow$ Dense$(32, \text{ReLU}) \rightarrow$ Dense$(1, \text{Sigmoid})$.
- **Loss & Optimization:** Binary Cross-Entropy Loss, Adam optimizer ($\alpha = 10^{-3}$), batch size 512, Early Stopping (patience $= 5$).

Figure 4 plots the learning curves, demonstrating steady convergence without overfitting.

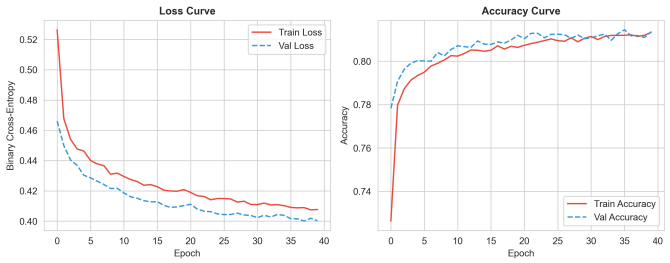


Fig. 4. Neural Network training and validation loss and accuracy curves.

VII. Model Selection and Comparison Analysis

A. Quantitative Performance Summary

Table I summarizes the test performance across all models.

TABLE I
PERFORMANCE COMPARISON OF ALL MODELS ON TEST SET

Model	Accuracy	Precision	Recall	F1-Score	AUC
Neural Network	**0.8151**	0.7197	0.5482	**0.6223**	**0.8585**
Decision Tree	0.8039	**0.7305**	0.4668	0.5696	0.8305
Logistic Regression	0.7752	0.7000	0.3345	0.4527	0.7971
KNN ($k = 5$)	0.7737	0.6087	0.5203	0.5611	0.7841
Gaussian NB	0.4181	0.3199	**0.9711**	0.4813	0.7388

B. Visual Performance Comparisons

Figure 5 compares prediction accuracy, while Figure 6 presents the Precision, Recall, and F1-Score trade-offs.

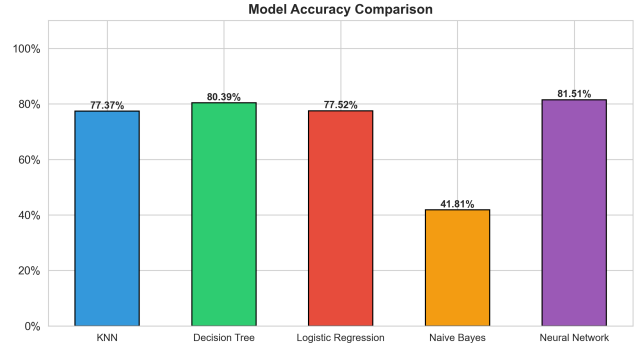


Fig. 5. Overall classification accuracy comparison across all models.

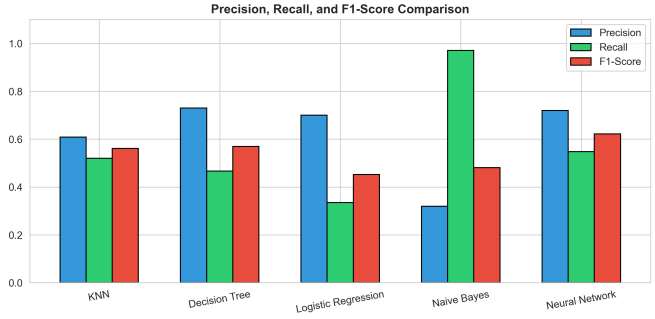


Fig. 6. Precision, Recall, and F1-Score breakdown for all models.

C. Error Analysis and Discrimination Capability

Figure 7 presents the confusion matrices for all five classifiers, highlighting the distribution of True Positives, False Positives, True Negatives, and False Negatives.

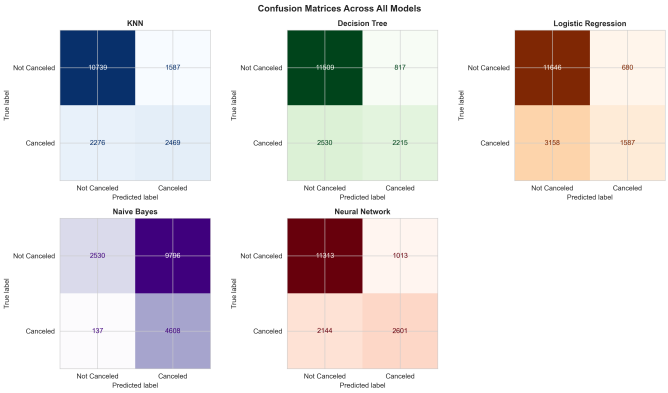


Fig. 7. Confusion matrices for all five classifiers on the test partition.

Figure 8 displays the Receiver Operating Characteristic (ROC) curves. The Deep Neural Network achieves the highest Area Under the Curve (AUC $= 0.8585$), demonstrating superior discriminative power across all decision thresholds.

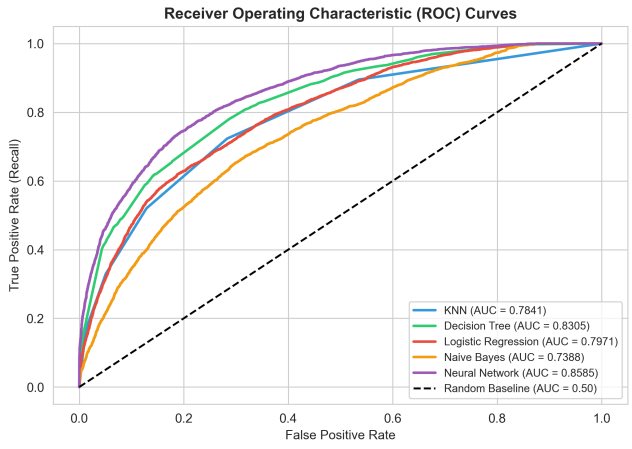


Fig. 8. Receiver Operating Characteristic (ROC) curves and AUC comparison.

VIII. Unsupervised Learning (K-Means Clustering)

To explore the intrinsic geometry of the dataset without supervision, we removed the target label `is_canceled` and applied K-Means clustering.

A. Elbow Method for Optimal k

We evaluated Within-Cluster Sum of Squares (Inertia) over $k \in [1, 10]$:

$$\text{WCSS} = \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|^2 \tag{6}$$

Figure 9 displays the Elbow curve, showing a sharp point of maximum curvature at $k = 2$, confirming that the feature space naturally partitions into two primary clusters.

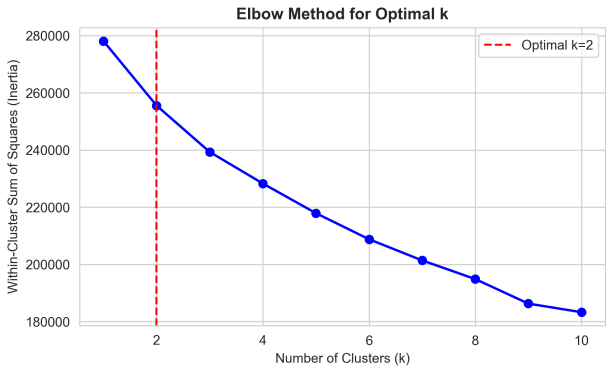


Fig. 9. Elbow Method plot evaluating WCSS (Inertia) for $k \in [1, 10]$.

B. PCA 2-D Cluster Projection

Using Principal Component Analysis (PCA) for dimensionality reduction, Figure 10 visualizes the K-Means clusters ($k = 2$) against actual ground truth cancellation labels.

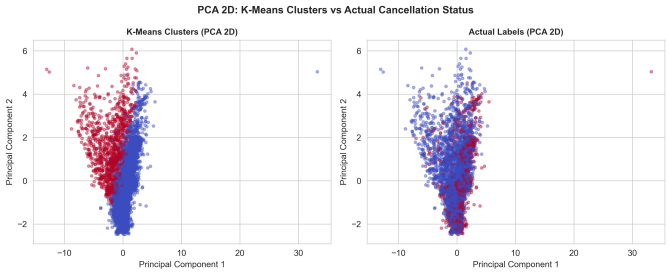


Fig. 10. PCA 2-D projection comparing unsupervised K-Means clusters ($k = 2$) against ground truth cancellation labels.

The strong spatial overlap between unsupervised clusters and true cancellation labels confirms that the dataset contains distinct behavioral clusters linked to cancellation risk.

IX. Conclusion

A. Best Performing Model

The **Deep Neural Network** is the best performing model, achieving the highest Accuracy (81.51%), F1-Score (0.6223), and AUC-ROC (0.8585). It effectively captures complex, non-linear interactions across mixed tabular features. The **Decision Tree** also demonstrated strong competitive performance (80.39% Accuracy, 0.7305 Precision).

B. Analysis of Performance Bottlenecks

- **Gaussian Naive Bayes:** Underperformed severely (Acc $= 41.81\%$) due to the strong violation of feature independence (e.g., correlations between deposit type, previous cancellations, and lead time).
- **Logistic Regression:** Showed limited recall (33.45%) because a linear hyperplane cannot adequately separate non-linear customer behaviors.

C. Challenges Encountered

- **Data Leakage:** Detecting and eliminating `reservation_status` was essential to ensure the model makes valid pre-arrival predictions.
- **High-Cardinality Nominals:** Managing `country` (178 levels) required careful dimensionality management.
- **Class Imbalance:** Addressed using stratified sampling to prevent minority class degradation.

D. Future Work

Future enhancements will explore gradient boosting tree ensembles (XGBoost, LightGBM, CatBoost), SMOTE oversampling to boost positive class recall, and SHAP explainability for operational deployment in hotel revenue management platforms.

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